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Sustainable Leadership and Conflict Management: Insights from Greece's Public Sector

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Abstract: This study investigates the relationship between sustainable leadership styles and conflict management strategies within the context of Greek Public Sector. Specifically, it examines how collaborative, transformational, and authoritarian leadership styles impact workplace conflict resolution. The research adopts a case study methodology, focusing on Departments of Public Works in Greece, where data were collected through questionnaires. The analysis involved quantitative methods, including exploratory factor analysis (EFA), to examine the relationship between leadership styles and conflict management techniques. Results indicate that collaborative leadership is strongly associated with higher employee satisfaction and more effective conflict resolution, particularly in organizations with flat hierarchical structures. Transformational leadership fosters trust and open communication, which further enhance conflict resolution. On the other hand, authoritarian leadership styles correlate with increased workplace tension, lower satisfaction, and less effective conflict management, especially in high power-distance environments. The study also highlights cultural factors, such as the Greek emphasis on interpersonal relationships, as critical influences on leadership effectiveness. These findings underline the need for culturally adaptive and sustainable leadership strategies and provide practical recommendations for promoting harmony and productivity in Greek organizations.



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1. Introduction

Conflict is an inevitable aspect of the workplace due to the diverse differences among leadership styles and conflict management are critical elements influencing organizational effectiveness. While existing studies have examined these dimensions extensively, the interplay between hierarchical perceptions and the practical application of leadership strategies remains underexplored, particularly in the context of public sector organizations. In Greece, the deeply hierarchical structure of the public sector creates a unique setting to investigate how supervisors and employees perceive leadership styles and conflict management techniques differently [1,2].

Using a quantitative, cross-sectional methodology, the study employed structured questionnaires distributed to supervisors and employees in the Central Macedonia Region of Greece. The analysis incorporates exploratory factor analysis (EFA) and regression

techniques to identify patterns and correlations, providing a robust statistical foundation for the findings.

The Greek public sector, characterized by rigid structures and significant cultural influences, presents a valuable case for studying these dynamics. Leadership practices in such environments often reflect broader societal norms, such as high power distance and emphasis on interpersonal relationships. Understanding how these cultural dimensions intersect with organizational hierarchies can reveal critical insights into the challenges and opportunities for effective conflict management [3].

The review of the existing literature highlights a notable gap in understanding the intersection of leadership styles and conflict management within the Greek public sector. This gap, coupled with the unique hierarchical structure of Greek organizations, guided the formulation of the study's research questions and methodology. By focusing on the interplay between cultural values, hierarchical perceptions, and leadership styles, this research fills a critical gap in the literature. It aims to not only expand theoretical knowledge but also provide actionable insights for fostering effective leadership and conflict management in public sector organizations [4,5].

2. Literature Review

Exploring how leadership styles interact with conflict management is essential to understanding their impact on organizational dynamics. This section delves into the core leadership styles and their role in addressing workplace conflicts.

2.1. Leadership Styles and Conflict Resolution

The leadership style that leaders will adopt is the key to the effectiveness of modern organizations. Leadership can motivate employees towards organizational objectives [6]. The ability to manage conflict is an essential skill for leaders, enabling them to cultivate a workplace that supports productivity and collaboration. A key factor is the leadership style adopted by leaders and its impact on conflict management approaches. Studies have shown that different leadership styles, such as team-oriented or authoritarian, can significantly influence the way in which conflicts are identified, addressed and ultimately resolved within organization [7]. Wang's research (2023) indicates that main factors for organizational sustainability are conflict management and employee performance [8]. The most frequently used conflict resolution techniques are: the technique of avoidance, the technique of compromise, the use of power, the technique of "organizing", the technique of confrontation and the mitigation [9,10].

In most of the existing research there is a theoretical assumption that the leader's approaches in dealing with a conflict is generally well communicated and understood by employees. But in real life there is a need by the leaders to be aware of how their conflict management techniques are received by the employees [11,12].

Different perceptions regarding these conflict resolutions approaches and leadership style can impact the dynamics of the workplace, the performance of the employees and their job satisfaction. Hussein et al. (2022) suggest that employees' views on their manager's leadership style, along with their self-confidence in achieving organizational objectives, play a crucial role. Building on these insights, the integration of sustainability into leadership practices offers an innovative pathway to address such challenges. Sustainable leadership extends beyond traditional organizational goals, emphasizing the creation of a workplace culture that nurtures trust, collaboration, and long-term commitment. It prioritizes employee development and empowerment, fostering an environment where individuals feel valued and motivated to contribute meaningfully. In hierarchical organizations, this approach bridges gaps between leadership and staff, enabling a shared

vision that aligns individual aspirations with organizational objectives. By embedding sustainability principles into leadership strategies, managers can cultivate resilience and adaptability, essential traits for navigating complex workplace dynamics. Additionally, the perceived quality of the manager-employee relationship, especially in terms of support provided, seems to influence performance outcomes [13–15].

Richmond et al. (1983) found that subordinates who perceived their supervisor as adopting a slightly more employee-focused leadership style (consultative or participative) reported greater satisfaction and solidarity, along with reduced communication anxiety [16,17].

In particular, in public sector where it is of great importance the service quality and the accomplishment of their aim, the leaders need to find ways to motivate employees. [18,19]. Asif and Rathore (2021) argues that there is a correlation between poor human resource decision by leaders and poor performance in service delivery [20].

Moreover, conflicts in the public sector are comparatively complicated to address because of the limitations that derive from the hierarchical and bureaucratic structure and from the wide range of the stakeholders (citizens, employees, elected officials, organized group with multiple interests). Conflict management involves integrating various factors that contribute to effectively resolving conflicts [21–23].

Ultimately, effective management relies on a well-functioning communication process, as managerial decisions, their outcomes, and an organization's vision become meaningful only when conveyed to employees and other stakeholders [24]. Thus, managing conflicts effectively fosters cooperation, builds mutual trust among organizational members, and strengthens their relationships [25].

Backhaus and Vogel (2022) argue that most scholars agree that leadership play a crucial role in public organization but it is not obvious which styles of leadership lead to certain outcomes and their impact [26]. Regarding leadership style, cooperative leadership style emphasizes teamwork, cooperation and joint decision making among team members. This leadership style is characterized by open communication and mutual respect [27]. In conflict management, collaborative leadership emphasizes problem solving and finding mutually beneficial solutions for all parties involved. Leaders who use a collaborative approach to conflict management encourage open communication and active listening among team members [28]. Collaborative leaders use conflict management techniques such as compromise and “organize” to resolve conflicts and achieve mutually acceptable outcomes [29,30]. The collaborative leader focuses on the emotional state of the employees and is therefore driven to develop empathy, which is considered a fundamental skill for this leadership style. This specific leadership style is very necessary mainly to avoid conflicts and tensions within the team in stressful conditions, where motivation and the strengthening of relationships and team coordination are necessary [31,32].

Transformational leadership is a leadership style that emphasizes change and transformation. Leaders who adopt this approach focus on building strong relationships with their followers and strive to inspire their followers to achieve their full potential and work towards a common goal [33].

This leadership style has been shown to be a positive influence on conflict management techniques. Incorporating principles of sustainability into leadership approaches has further demonstrated its potential to enhance conflict management strategies. Sustainable leadership emphasizes long-term solutions that prioritize organizational well-being and employee satisfaction. Research highlights that collaborative and transformational leadership styles, when coupled with sustainable practices, foster trust, innovation, and open communication. These attributes not only improve conflict resolution but also contribute to a harmonious and productive work environment. Transformational leaders leverage

their charisma and persuasive abilities to inspire trust and motivate their teams, fostering a collaborative and values-driven organizational culture. According to Triantari (2020) the transformational leader can become the cause of transformation of the moral values of the subordinates in an organization [32]. He/She appears charismatic, activates and motivates the team, having the charisma or ability of persuasion to ensure a mutual relationship of trust among them [34,35]. Internal trust strengthens and affects positively organization's performance. As internal trust refers to the level of trust that employees have among them and in the leaders [36]. According to Du et al. (2021) transformational leadership demonstrates advantages over more traditional styles, in terms of meeting organizational objectives [37].

Authoritarian leadership is a leadership style in which the leader makes decisions and gives orders without input or feedback from subordinates. This leadership style is characterized by a rigid hierarchy, strict rules and procedures, focusing on achieving results rather than building relationships. In conflict management, authoritarian leaders tend to take a more competitive approach, where they use their power and authority to impose their decisions on others [38].

They may also use tactics such as threats, punishment, and coercion to resolve conflicts quickly and decisively. While authoritarian leadership may be effective in some short-term situations, such as in times of crisis or when there is a need for quick decision making, it may not be the best approach to managing conflicts in the long term. The authoritarian leader spreads insecurity and uncertainty to subordinates, communication is one-way, with frequent misunderstandings and miscommunications, he makes decisions alone, without considering his partners and deconstructs teamwork and cooperation [39].

A great deal of contemporary studies regarding leadership and management are influenced of the idea of sustainability. Sustainable leadership aims to a balanced development of economy, society and environment hoping to create long term value for all stakeholders [40]. Under this perspective of long-term wellbeing and value for the stakeholders, the concept of sustainable leadership is broader than the social and environmental sustainability [41]. Moreover, it involves building a strong team, long-term decisions, fostering innovation and providing products, services and solutions of high quality [42]. Employees, as also stakeholders, are priority to sustainable leadership, which aims to create a solid organizational culture. A strong organizational culture with values shared from all the stakeholders lead to resilience and long-term success [43]. Thus, according to Iqbal et al. (2020b) sustainable leaders emphasize in employees' development and as result employees support and embrace their leaders' sustainable behaviors [44].

Furthermore, the relationship between sustainable leadership and the leadership styles examined in this study can be further clarified. Collaborative and transformational leadership styles align closely with the principles of sustainable leadership. Collaborative leadership fosters an inclusive environment where shared decision-making and mutual trust contribute to a resilient organizational culture—key aspects of sustainability. Transformational leadership, on the other hand, inspires innovation, long-term vision, and personal development, elements that reinforce sustainable growth. In contrast, authoritarian leadership, which often limits participation and open communication, may hinder the development of a sustainable workplace culture. Therefore, understanding how these leadership styles support or challenge sustainable leadership principles is essential for promoting resilience, adaptability, and long-term organizational success [32].

Public leaders operating in complex service environments must navigate the challenge of balancing routine operational priorities with unexpected demands, especially within the context of limited resources and heightened performance expectations [32].

Getha-Taylor et al. (2011) argue that public leadership should be treated as a distinct field within leadership studies and prioritized in public administration research. They highlight that public managers operate under unique constraints and requirements compared to other leaders [45].

Furthermore, Orazi et al. (2013) argue that public sector leadership is an emerging field that requires tailored strategies for effective conflict resolution. Their findings suggest that transformational and transactional leadership styles can significantly influence conflict management outcomes [46].

The Khan et al. (2015) research also supported for the existence of a relationship between the leadership/managerial style and modes of conflict management [47].

Rainey (2009) contends that bureaucratic constraints and ambiguous goals in such settings hinder the full implementation of transformational leadership, as they complicate the process of establishing clear organizational visions. Dwivedi (2014) suggests that transformational leadership could be highly effective in public organizations due to their strong emphasis on service-oriented and community-focused missions. A transformational leader focuses on inspiring and motivating employees which aligns with the need of public service motivation. Researches consistently indicate a positive correlation between transformational leadership and employee performance in the public sector. A transformational leader focuses on inspiring and motivating employees which aligns with the need of public service motivation [48,49].

Transformational leaders exert a strong effect on the ways in which workers view their job. Existing research deepens the knowledge concerning the collaborative approaches of the transformational leader to conflicts [30,50,51].

2.2. Collaborative Leadership in Practice

As organizations evolve, collaborative leadership has emerged as a critical approach to fostering teamwork and resolving conflicts effectively. The following section highlights the principles and practices of this leadership style.

Modern leadership necessitates an approach that actively listens to employees, addresses their concerns, and simultaneously maintains focus on shared goals, fostering their achievement through collaboration. This approach aligns with the principles of collaborative leadership [52].

Van Wart (2013) identifies the emergence of a third model of public sector leadership, distinct from the traditional hierarchical and public choice models. This approach prioritizes collaborative processes aimed at achieving shared outcomes across agencies and sectors, while enhancing democratic accountability to promote responsiveness and inclusiveness. It is commonly referred to as facilitative leadership, collaborative leadership, or public values leadership. Concerning performance, collaborating leadership will promote organizational performance to be able to remain competitive and survive and it will impact to the whole organization [53,54].

Collaborative leaders focus on guiding and inspiring network participants rather than exerting control or providing direct instructions [55]. Collaborative leadership is not merely a process of developing the skills needed to foster collaboration; it also involves creating an environment where teams can cultivate the competencies and behaviors essential for effective collaboration, thereby supporting the organization's financial sustainability [52].

On the contrary, authoritarian leaders enforce strict hierarchical control, expecting subordinates to demonstrate submission, dependency, and obedience [56]. In hierarchical structures, subordinates tend to view obedience to the legitimate authority of authoritarian leadership as a necessity. Recent management studies indicate that it negatively impacts subordinate job satisfaction and trust in leadership [57]. Obi et al. (2021) in their research

argued that authoritarian leadership is found to be strongly and positively related to avoiding and forcing third-party behaviors. Authoritarian leaders may use personal power and coercive authority to opt not to intervene in followers' conflicts, as conflicts may not seem relevant [58].

Leadership fundamentally involves interaction and the influence of an individual on their group. Its effectiveness relies on the alignment between how leaders and followers perceive the world and the leadership process within it. Leaders cannot effectively convey their perspective on the current state and vision for a new reality to their followers if there is no shared understanding of fundamental societal dynamics [59].

Furthermore, bureaucracy under the lens of public administration plays a significant role in conflict management as it sets a solid framework with rules and regulations concerning conflict resolution. However, it can also set barriers with the lack of flexibility or the crucial role of the bureaucrat/leader, who is responsible to interpret the conflict management [60].

2.3. Cultural Influences on Leadership

Leadership styles are deeply influenced by cultural values and norms, shaping the approaches leaders take in managing teams and resolving conflicts. This section examines the interplay between cultural dimensions and leadership practices

Numerous studies have explored the relationship between culture and leadership styles, revealing that different cultural groups often hold distinct views on leadership. Bersin and Chamorro-Premuzic (2020) observe that the characteristics of successful leaders differ between developed and emerging economies. As a result, both individual and national culture shape the leadership style and conflict management strategies adopted by leaders. Leadership roles are influenced by societal values, structural norms, and communication styles, all of which guide conflict resolution approaches [61,62].

Leadership approaches vary across cultures, as leaders and their followers often have differing perspectives on the realities of their environment [58,63].

Power distance and the individualism–collectivism dimension play a significant role in influencing leadership style preferences. Cultures characterized by collectivism or high power distance tend to favor authoritarian leadership, whereas individualistic cultures or those with low power distance are more inclined toward participative leadership. For example, authoritarian leadership styles are still predominant in many Asian cultures. At the same time the leaders in these countries are expected to use avoiding or dominating conflict management techniques. Moreover, in hierarchical structures employees are not very much involved in decision making [64,65].

Backhaus and Vogel (2022) in their research shed a new light in how different cultures affect leadership. They argue that the outcomes of leadership in public sector in Anglo-American tradition are more beneficial than these in German and Scandinavian tradition. The public sector in German and Scandinavian countries is highly structured with rules and regulations that may lead leaders to rule-following and controlling leadership styles. On the other hand, there is more flexibility in Anglo-American cultures which provides efficient space for leader's managerial skills. Their conclusion is that transformational leadership is more likely to emerge in Anglo-American countries than in continental European countries [26,66,67].

There are few studies examining leadership styles within Greece's Public Sector, focusing mainly in health care sector and in education. Menon's research (2021) resolved that the bureaucratic nature of Greek educational system is one of the factors that prevent school leaders from being transformative [68]. In a similar research in Greek hospitals Goula et al. (2021) argue that the culture in these environments put severe limitations in employees'

participation in decision making due to the way the hospitals are structured following laws and procedures and a prevalent controlling system. Moreover, the research notices the tendency of Greek health care employees to act individually and not appreciate the value of teamwork [69]. The results from an empirical study from Bank Sector employees in Greece reveal that transformational and democratic management styles are more appropriate to cope the global financial market competition [70]. Furthermore a study in Greek tourism Services reveals a connection between strategic leadership and work commitment [71].

Also, Kaushal & Kwantes (2006) highlighted the role of culture in shaping conflict management and resolution behaviors, noting that individualism and collectivism significantly affect an individual's preferred style of resolving conflicts [72]. Moreover, recent research provides insights on the ways that culture influences leadership [73].

Across the public, private, and non-profit sectors, there is a growing shift away from rigid hierarchical leadership styles toward fostering collaboration and cooperation among employees to enhance productivity. While horizontal leadership is becoming more common in team settings, managers accustomed to traditional individual problem-solving methods may find it challenging to fully embrace this approach [32,74].

For this analysis we focused on three contents: organizational level, culture and leadership even if there are numerous contents to examine as well as a vast research in literature. Nevertheless, the reality in Greek public Sector as it is presented in relevant literature focuses in these elements. There are significant findings in existing research literature but they often lack deeper analysis [3–5,70,75]. The current paper aims to fill in this gap and addresses the following questions.

Moreover, existing literature suggests that organizational level, culture, and leadership are interdependent elements that fundamentally shape conflict management practices. The organizational level determines the structural framework within which leadership operates, influencing hierarchical relationships and decision-making processes. Cultural factors affect leadership expectations, trust dynamics, and conflict resolution preferences. Leadership, as the driving force of organizational behavior, is shaped by and simultaneously shapes both the organizational context and cultural environment. Thus, these three contents provide a comprehensive analytical framework, as they collectively determine how conflict management techniques are adopted and perceived. This integrated approach offers deeper insights than isolated examinations of other factors, such as individual competencies or policy frameworks, which often lack contextual relevance [5]. This comprehensive analytical framework not only addresses existing gaps in literature but also directly supports the research questions of this study by illustrating how organizational structure, cultural context, and leadership dynamics jointly influence conflict management approaches. Such an integrated perspective is essential for understanding the complexity of leadership practices within the Greek public sector [3].

Last but not least, the selection of organizational level, culture, and leadership as the key constructs for this study's analysis is based on their fundamental role in shaping leadership practices and conflict management strategies in hierarchical organizations. Previous research highlights the impact of organizational structure on leadership dynamics, where the level of authority within the organization determines how leadership is exercised and decisions are made. Similarly, culture influences leadership expectations, trust, and communication styles, all of which are critical in conflict resolution processes. Lastly, leadership, as the central driver of organizational behavior, both shapes and is shaped by the organizational structure and cultural context. These three factors are interdependent and provide a comprehensive framework for understanding how leadership styles and conflict management techniques are adopted and perceived in the public sector, especially in Greece's unique institutional and cultural environment [38].

2.4. Research Questions

Exploring the relationship between leadership styles and conflict management techniques is essential for understanding how these factors influence workplace dynamics. Particularly in the Greek public sector, where hierarchical structures and cultural values are deeply ingrained, it is crucial to examine how these variables interact. This study aims to investigate these interactions by focusing on the perceptions of different hierarchical levels within public organizations.

The following research questions aim to address the identified gaps in the literature by focusing on the perceptions of leadership styles and conflict management techniques within the Greek public sector. Perception of leadership is a multifactorial issue and deals with the interpretation of leadership by individuals and the society. Leadership perception is different across countries, societies and individuals. It is a subjective concept that also may differ from the act of leadership both in theoretical approach and in practical application. However, there is a connection between the two concepts as the way leadership is perceived affects the leader and the outcomes but also the way the leader applies leadership influences employee's perception. By examining these issues across hierarchical levels, the study seeks to provide a comprehensive understanding of the interplay between leadership practices and organizational dynamics in a culturally unique setting.

Differences in perceptions regarding conflict management techniques can significantly impact organizational cohesion and efficiency. Existing literature highlights that leaders and subordinates often have divergent views on the methods employed to resolve conflicts. This question addresses whether such differences exist in the Greek public sector, shedding light on the role of hierarchical positions in shaping these perceptions [3].

RQ1: Do supervisors and employees differ in their perceptions of the conflict management techniques employed by leaders in the public sector?

Leadership styles are central to shaping workplace dynamics, and perceptions of these styles can vary between supervisors and employees. Understanding these differences is critical for aligning leadership practices with organizational goals. This question examines whether supervisors and employees perceive leadership styles differently within the public sector context [5].

RQ2: Do supervisors and employees differ in their perceptions of the leadership styles employed by leaders in the public sector?

The link between leadership characteristics and conflict management techniques is pivotal in understanding how leaders influence workplace harmony. Prior research suggests that certain leadership traits can predict the adoption of specific conflict resolution strategies. This question further explores whether leadership traits can predict the specific conflict resolution strategies employed by leaders in the Greek public sector [4].

RQ3: Do leadership characteristics predict the conflict management techniques used by leaders in the public sector?

3. Methodology

3.1. Research Planning

For the purpose of the present study a quantitative, primary, cross-sectional, correlational research was conducted with the use of questionnaire, which reliability was tested by the use of Cronbach Alpha coefficient [76]. Primary, cross-sectional, quantitative research was considered the appropriate plan in order to ascertain in the current time period the opinions of public sector managers and employees on measurable concepts [77], which are the frequency of application of conflict management techniques and the leadership style which leaders apply, concepts that formed the dependent variables. Employees are seen as a valuable source of information because they have a good view on internal processes [78,79].

The existence of specific research questions required data that derive from certain source and couldn't be retrieved from other sources. Besides, it was crucial to understand behaviors, characteristics, styles etc. at a current time period. The quantitative method was possible because of the availability of the data and the capability of implementing measurements with the use of the measurable concepts. Moreover, it was needed to examine the relationships between two different variables [80].

Consequently, this approach provides a measurable, statistically validated understanding regarding the interaction of leadership styles and conflict management techniques enhancing the reliability and accuracy of its findings. Measurable concepts were measured using Likert scale questionnaires which were tested for reliability and validity [81,82]. Quantitative research was used to test the effect of independent variables on dependent variables using statistical techniques on numerical data. Holding a leading position was considered the independent variable of the research in the Public Sector with 2 levels (supervisor, employee). Leadership styles were considered as independent variables when investigating their possible relationship with conflict management techniques [83].

3.2. Process

Data collection process began with two types of questionnaires (one addressing to the supervisors, the second to the employees) that were sent to the public servants working at the Departments of Public Works in Region of Central Macedonia, Greece. In the introductory note of the questionnaires, it was emphasized that anonymity is fully respected, as well as the protection of personal data by the University of Western Macedonia, and also that the answers of the participants will be used only for scientific purposes.

To ensure the validity and relevance of the survey instrument, we engaged in a comprehensive validation process. Expert reviews were conducted by professionals with expertise in leadership, conflict management, and organizational behavior. These experts assessed the clarity, relevance, and content validity of the questionnaire items to ensure they accurately measured the intended constructs. Additionally, a pilot test was conducted with 30 participants, representative of the study's target population, to evaluate the instrument's clarity, time requirements, and potential issues with item interpretation. Based on the feedback from the pilot test, minor revisions were made to improve the clarity and wording of some questions.

In the main data collection process, no incentives were provided to participants. The surveys were distributed directly to employees and supervisors within the Departments of Public Works in Greece, ensuring that the sample included only the relevant target group. While convenience sampling was used, which limits the generalizability of the results, efforts were made to mitigate biases by ensuring that the sample consisted of appropriate participants who met the study's criteria.

In addition, it was emphasized that the survey is voluntary and participation in it will take place with the consent of the participants, as well as the analysis of the responses in accordance with Data Protection (GDPR), where the possibility of withdrawal and access to the data for corrections and additions is also possible as restriction of data processing. Participants had the right to contact the Personal Data Protection Authority if they found that the processing of their personal data violates the GDPR.

The researcher communicated her personal e-mail to the participants in case they want to contact for any reason. The questionnaires were sent to the Public Works Departments of the 7 Regional Prefectures and the 38 Municipalities of Region of Central Macedonia in Greece. The distribution was made electronically through the Human Resources Department of each service, after the consent of the Managers of each Department. An email with

google forms application was sent to the e-mails of all the employees/supervisors of each Department. The necessary moral and ethical issues of a scientific research were observed.

3.3. Questionnaire

The questionnaire was selected as the most suitable research tool, as it enables a large sample to be approached, providing a general picture of the problem to be investigated that is difficult to capture with other research techniques. So, a structured questionnaire was designed which is divided into 3 sections [84]:

(A) Demographic data: The section consists of 5 closed-ended questions regarding gender, age, education level, years of service and position of responsibility in the Public Sector.

(B) Conflict Management Techniques: A questionnaire of 5 questions on a Likert scale from 1–5 (1 = Never, 2 = Rarely, 3 = Sometimes, 4 = Often, 5 = Very often) was used regarding the frequency of application of the following conflict management techniques:

1. The technique of avoidance, which refers to the separation of conflicting sides, defining tasks and postponing problem solving [9,85].
2. The compromise technique in which the two parties are asked to find a common solution, without benefiting either side [9,85].
3. The confrontation technique where the two sides come into contact, at the initiative of the manager, to eliminate misunderstandings [86].
4. The use of authority in which the manager orders his subordinates to stop the conflict, when it is not possible to find a solution [9,85].
5. The organizing technique where the nature of the conflict is analyzed to decide the most appropriate method [9].

The questionnaire is considered valid through content validity, meaning that the following techniques cover the possible spectrum of conflict management techniques [87].

(C) Leadership style: A questionnaire of 16 questions on a Likert scale of 1–5 (1 = Very little, 2 = A little, 3 = Neither a little, nor much, 4 = A lot, 5 = Very much) was used, regarding 3 leadership styles that emerged with the Factor analysis technique, with Equamax rotation with the eigenvalue criterion, proving conceptual construct validity [88]. The data were ideal for conducting the procedure ($KMO = 0.892 > 0.800$), with the 3 factors explaining 75.86% of the total variance (Table 1). Factors were tested for reliability using the Cronbach Alpha coefficient, which measures internal consistency reliability with values greater than or equal to 0.7 considered satisfactory [89].

These three leadership styles are considered to adequately cover the spectrum of leadership styles [70].

To assess the reliability of the survey instrument, we calculated Cronbach's alpha for each of the three factors identified in the exploratory factor analysis (EFA). The results showed high internal consistency: Factor 1-Collaborative Leadership- ($\alpha = 0.947$), Factor 2-Transformational Leadership- ($\alpha = 0.907$), and Factor 3- Authoritarian Leadership- ($\alpha = 0.800$). These values indicate that the instrument is highly reliable for measuring the intended constructs. Additionally, Eigenvalue values for the factors were 6.87, 3.38, and 1.89, respectively, supporting the validity of the factor structure. The combination of these results ensures that the survey instrument is robust and suitable for analyzing the relationships between leadership styles and conflict management techniques.

The leadership styles that emerged are:

1. Collaborative: The factor includes 7 questions such as e.g., "Creating positive communication", "Confidence and strengthening your colleagues" and "Team spirit", explains 42.92% of the total variance and showed excellent reliability ($\alpha = 0.947$).

2. Transformative: The factor includes 3 questions such as e.g., “Encouraging, encouraging and animating colleagues” and “Ability to change and improve work”, explains 21.13% of the total variance and showed excellent reliability ($\alpha = 0.907$).
3. Autocratic: The factor includes 6 questions such as e.g., “Authoritarianism”, “Strictness” and “Discipline”, explains 11.81% of the total variance and showed high reliability ($\alpha = 0.800$).

Table 1. Results of Factor Analysis with Equamax Rotation for Leader Characteristics.

Questions (KMO = 0.892)	Variable		
	Collaborative	Transformational	Authoritarian
16. Emotional Intelligence and empathy	0.916		
13. Create Positive Communication	0.910		
10. Persuasion	0.909		
5. Self-confidence and colleagues strengthening	0.886		
3. Belief in values	0.839		
14. Team spirit	0.838		
1. Vision and need for achievement	0.719		
11. Communication		0.920	
12. Encouraging, Motivating and Inspiring colleagues		0.866	
15. Ability to change and job development skills		0.779	
8. Authoritarianism			0.656
9. Severity			0.584
6. Obsession			0.742
2. Need of social recognition			0.665
4. Need for power projection			0.534
7. Discipline			0.469
Variance (%)	42.92%	21.13%	11.81%
Eigenvalue	6.87	3.38	1.89
Cronbach Alpha	0.947	0.907	0.800 *

* $\alpha = 0.1$.

To assess the potential impact of common method variance (CMV) on the results, we conducted the Harman’s Single Factor Test on the survey data. The results revealed that the largest factor accounted for 40% of the total variance, which is well below the 50% threshold that indicates significant common method bias. This suggests that common method variance is not a major concern in this study. Furthermore, to minimize the risk of CMV, we ensured that data were collected from multiple respondent groups, including both employees and supervisors within the Departments of Public Works in Greece. This approach helped to reduce the likelihood of common method bias by gathering responses from diverse sources, contributing to the robustness of the findings. By implementing these measures, the potential impact of CMV on the study’s results has been effectively mitigated.

3.4. Population-Sample

The population of the research is considered to be all civil servants of the Public Works Departments of the 7 Regional Prefectures and the 38 Municipalities of the Region of Central Macedonia in Greece. The sample was collected using convenience sampling

as the participants were approached through the google forms application from the managers or the human resources of the departments to which the researcher had contact access. Convenience sampling was considered the appropriate method for practical and strategic reasons.

This method has a simplicity in implementation as the study was addressing in a specific target group. Besides the group was easily accessible and therefore the data collection took place within the required time period. Convenience sampling is extensively used in real-world research, especially in localized studies like the present. Generalization of the study is applicable to population with similar characteristics to the sample because convenience sampling deals with a certain group and not random selection. But correlation research offers the opportunity to generate the research for further experimental studies [87,90].

The sample of the 7 Prefectures and the 38 Municipalities represented a diverse range of categories. It included Municipalities with large/small population, metropolitan and urban areas as well as agricultural municipalities. Some of them are located by the sea, others on the mountains. Therefore, they are fulfilling all the possible criteria.

The sample consisted of 186 individuals, of which 132 were employees (out of a total of approximately 380 employees, i.e., approximately 35% of the employees participated in the survey and constitute 71% of the survey participants) and 54 supervisors (out of a total of approximately 136 supervisors, i.e., approximately 38% participated in the survey and constitute 29% of survey population) and was almost equally distributed by gender (54.6% female, 45.4% male).

The majority of participants were 40–59 years old (90.3%), with a university or post-graduate education level (89.2%) and more than 10 years of service in the Public Sector (89.2%) (Table 2).

Table 2. Demographic and Occupational Data.

Question	Categories	N	%
Gender	Female	101	54.6
	Male	84	45.4
	Other	0	0
Age	25–39	5	2.7
	40–49	73	39.2
	50–59	95	51.1
	≥60	13	7.0
Years of service in Public Sector	1–10	20	10.8
	11–20	66	35.4
	>20	100	53.8
Level of Education	Basic	12	6.5
	Bachelor	73	39.2
	Master Degree	93	50.0
	PHD	8	4.3
Role in Public Sector	Manager	54	29.0
	Employee	132	71.0

3.5. Data Analysis

Data analysis was performed at a significance level of 5% in IBM SPSS 26 software. Factors were tested for normality using the Shapiro Wilk test [90,91]. Shapiro Wilk test is reliable even in small sample sizes and commonly used in similar studies. Due to non-existence of a normal distribution ($p \leq 0.028$, Table 3), non-parametric tests were implemented. Mann-Whitney U test replaces the t-test for comparing two groups and to analyze relationships between variables, Spearman's correlation replaces Pearson's correlation. Therefore, the non-parametric Mann-Whitney U test was used to compare the perceptions between supervisors and employees on the frequency of application of certain conflict management techniques and leadership styles followed by leaders. Values were presented with median and interquartile range [92].

These tests are addressing the first and second research questions

(1) Do supervisors' and employees' opinions about the conflict management techniques used by leaders in the Public Sector differ?

(2) Do supervisors and employees' opinions differ on the leadership style employed by leaders in the Public Sector?

Regarding the perceptions of employees and leaders concerning the leadership style and the use of conflict management techniques. Moreover, they provide statistical results whether significant differences between these perceptions exist. Medians and IQRs offers a clear summary of the central tendency and dispersion of responses within each group, facilitating the comparison of the results.

Similarly, to address the third research question "Do leadership characteristics predict the conflict management techniques used by leaders in the Public Sector?" the non-parametric Spearman correlation coefficient was used to check the correlation of leadership styles with conflict management techniques. With the Spearman's rho, the study identified which leadership characteristics were significantly correlated with specific [93].

Finally, to investigate the predictive power of leadership styles on conflict management techniques, multiple linear regression models were used. Leadership characteristics that were related to conflict management techniques in the bivariate analysis were the independent variables while the conflict management techniques defined as the depended variables [94]. This analysis provides the results on how the leadership style could predict the implementation of a particular conflict management technique.

Table 3. Normality tests using Shapiro Wilk test.

Variable	W	df	p
Avoidance	0.916	186	<0.001
Compromise	0.891	186	<0.001
Confrontation	0.909	186	<0.001
Authority	0.897	186	<0.001
Organising	0.904	186	<0.001
Collaborative	0.950	186	<0.001
Transformational	0.934	185	<0.001
Authoritarian	0.984	186	0.028

The combined use of Spearman's correlation and multiple linear regression analyses provides an effective way to examine the research question. The first step is the correlation analysis about the possible relationships between leadership styles and conflict manage-

ment techniques. The final step is the regression analysis which evaluates the prediction of these techniques according the leadership style.

4. Results

4.1. Comparison of Views on Conflict Resolution Techniques

Regarding the first research question “Do supervisors’ and employees’ opinions about the conflict management techniques used by leaders in the Public Sector differ? and according Table 4, supervisors felt to a higher degree compared to employees that the leader uses compromise (MdnSupervisors = 3.70 vs. MdnEmployees = 2.89, $p < 0.001$), confrontation (MdnSupervisors = 3.06 vs. MdnEmployees = 2.60, $p = 0.008$) and organizing technique (MdnSupervisors = 3.72 vs. MdnEmployees = 2.50, $p < 0.001$) to deal with conflicts while employees felt that the leader uses the avoidance technique more (MdnSupervisors = 2.00 vs. MdnEmployees = 3.22, $p < 0.001$) (Table 4).

Therefore, it is obvious that the supervisors’ and employees’ perception about the conflict management techniques used by the leaders in the Public Sector do differ. Supervisors supported that leaders use compromise, confrontation and organizing techniques to manage conflicts while employees supported that the leaders use the technique of avoidance.

Table 4. Comparison of opinions regarding the level (position of responsibility) in Public Sector.

Variable	Supervisor	Employee	U	p
Avoidance	2.00 (1.02)	3.22 (1.18)	2200.5	<0.001
Compromise	3.70 (0.72)	2.89 (1.09)	2002.0	<0.001
Confrontation	3.06 (0.92)	2.60 (1.20)	2702.5	0.008
Power	2.32 (1.03)	2.70 (1.28)	2987.5	0.075
Organising	3.72 (1.04)	2.50 (1.25)	1637.5	<0.001
Collaborative	4.14 (0.71)	3.07 (1.57)	1410.0	<0.001
1. Vision and need of achievement	4.00 (1.00)	3.00 (2.00)	2604.5	0.003
3. Belief in values	4.00 (1.00)	3.00 (2.00)	1762.0	<0.001
5. Self-confidence and colleagues strengthening	4.00 (0.50)	3.00 (2.00)	1622.0	<0.001
10. Persuasion	4.00 (1.00)	3.00 (2.00)	1806.0	<0.001
13. Create positive communication	4.00 (1.00)	3.00 (2.00)	1578.0	<0.001
14. Team Spirit	4.00 (1.00)	3.00 (2.00)	1563.5	<0.001
16. Emotional Intelligence and empathy	4.50 (1.00)	3.00 (2.00)	1716.5	<0.001
Transformational	4.00 (0.75)	2.33 (1.67)	959.5	<0.001
11. Communication	4.00 (1.00)	2.00 (2.00)	1012.0	<0.001
12. Encouraging, motivating and inspiring colleagues	4.00 (1.00)	2.00 (2.00)	866.5	<0.001
15. Ability to change and job development skills	4.00 (0.25)	3.00 (2.00)	1676.0	<0.001
Authoritarian	2.30 (0.85)	3.20 (1.40)	1995.5	<0.001
2. Need of social recognition	3.00 (1.25)	3.00 (2.00)	3413.5	0.641
6. Obsession	2.00 (2.00)	3.00 (2.00)	2390.0	<0.001
7. Discipline	3.00 (1.00)	3.00 (2.00)	3552.5	0.972
8. Authoritarianism	1.00 (1.00)	3.00 (2.00)	1159.0	<0.001
9. Severity	2.00 (2.00)	3.00 (2.00)	1870.5	<0.001

4.2. Comparison of Views on Leadership Style

Regarding the second question, Do supervisors and employees opinions differ on the leadership style employed by leaders in the Public Sector? and according Table 4, supervisors felt to a greater extent compared to employees that the leader practices collaborative leadership (MdnSupervisors = 4.14 vs. MdnEmployees = 3.07, $p < 0.001$) and specifically that he/she has a vision (MdnSupervisors = 4.00 vs. MdnEmployees = 3.00, $p = 0.003$), belief in values, persuasion, team spirit, that he/she strengthens their colleagues, that he/she creates positive communication (MdnSupervisors = 4.00 vs. MdnEmployees = 3.00,

$p < 0.001$) and that he/she has empathy (MdnSupervisors = 4.50 vs. MdnEmployees = 3.00, $p < 0.001$).

In addition, supervisors agreed more than employees that the leader applies transformational leadership (MdnSupervisors = 4.00 vs. MdnEmployees = 2.33, $p < 0.001$) and specifically that he/she is more communicative, that he/she encourages and empowers his/her colleagues (MdnSupervisors = 4.00 vs. MdnEmployees = 2.00), $p < 0.001$) and that he/she is capable of changing employees, improving them at work (MdnSupervisors = 4.00 vs. MdnEmployees = 3.00, $p < 0.001$). Finally, the employees considered to a greater extent than the superiors that the leader applies more of an authoritarian leadership (MdnSupervisors = 2.30 vs. MdnEmployees = 3.20, $p < 0.001$) which is expressed through obsession, strictness (MdnSupervisors = 2.00 vs. MdnEmployees = 3.00, $p < 0.001$) and authoritarianism (MdnSupervisors = 1.00 vs. MdnEmployees = 3.00, $p < 0.001$) (Table 4).

Therefore, it is obvious that the supervisors' and employees' perception about the leadership style employed the leaders in the Public Sector do differ. Supervisors supported that leaders employ collaborative leadership style which is expressed through vision and need for achievement, belief in values, self-confidence and colleagues strengthening, persuasion, positive communication, team spirit, emotional intelligence and empathy. Supervisors also believe that the leaders employ transformational leadership style which is expressed with communication, encouraging, motivation and inspiring colleagues, ability to change and job development skills. On the contrary employees supported that leaders employ authoritarian leadership style which is expressed with need for social recognition, obsession, discipline, authoritarianism and severity.

4.3. Leadership Style as a Predictive Factor of Techniques

Regarding the third research question, "Do leadership characteristics predict the conflict management techniques used by leaders in the Public Sector?" and according Table 5 and Figure 1, and Table 6 and Figure 2, the compromise technique was related to collaborative leadership ($\rho(186) = 0.447$, $p < 0.001$) and transformational leadership ($\rho(185) = 0.195$, $p < 0.01$), however, only collaborative leadership emerged as a predictive factor of compromise leadership ($\beta = 0.436$, $p < 0.001$), with the multiple linear model explaining 20.7% of the variance of the dependent variable. The confrontation technique was related to collaborative leadership ($\rho(186) = 0.442$, $p < 0.001$) and authoritarian leadership ($\rho(186) = 0.295$, $p < 0.001$), but only collaborative leadership was a predictive factor ($\beta = 0.421$, $p < 0.001$), of the comparison with the multiple linear models having a moderate degree of fit ($R^2 = 22.2\%$). Organizing technique was related to collaborative ($\rho(186) = 0.633$, $p < 0.001$), transformational ($\rho(185) = 0.280$, $p < 0.001$), and authoritarian leadership ($\rho(186) = 0.227$, $p < 0.01$), but only collaborative leadership was a predictive factor of organizational technique ($\beta = 0.681$, $p < 0.001$), with the multiple linear models explaining 43.4% of the variance of the dependent variable.

Table 5. Correlations of leadership styles with conflict management techniques.

Dependent	Collaborative	Transformational	Authoritarian
Avoidance	−0.154 *	−0.146 *	0.140
Compromise	0.447 ***	0.195 **	0.124
Confrontation	0.442 ***	0.126	0.295 ***
Power	−0.062	0.144	0.096
Organising	0.633 ***	0.280 ***	0.227 **

* $\alpha = 0.1$, ** $\alpha = 0.05$, *** $\alpha = 0.01$.

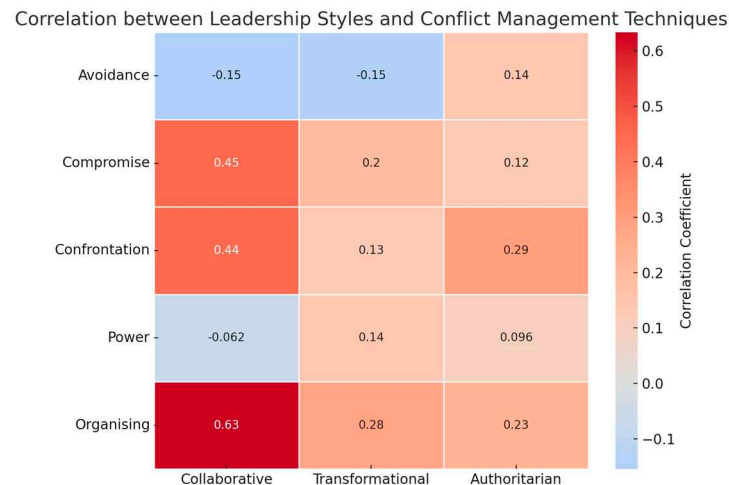


Figure 1. Correlation between Leadership Styles and Conflict Management Techniques.

Table 6. Multiple linear regression models.

Dependent	Collaborative	Transformational	Authoritarian	df1	df2	F	R ²	p
Avoidance	−0.101	−0.119	-	2	182	2.764	2.9%	0.066
Compromise	0.436 **	0.067	-	2	182	23.761	20.7%	<0.001
Confrontation	0.421 **	-	0.078	2	183	26.058	22.2%	<0.001
Organising	0.681 **	0.106	−0.114	3	181	46.229	43.4%	<0.001

** $\alpha = 0.05$.



Figure 2. Multiple linear Regression Models.

The results indicate that the leader who employs collaborative or transformational leadership style is predicted to use compromise, confrontation and organizing technique to manage conflict. Moreover, the prediction for this leader regarding the technique of avoidance is that he/she will not use it.

5. Discussion

5.1. Summary of Findings

Scholars in leadership literature argue that conflict management is a determining element of the leadership style followed by a leader or an organization. It is interesting

though, to determine to what extent conflict management can justify a leadership style or be justified by this, as we must consider other factors involved in its application.

As it was presented in Literature Review the three contents that we focus in this analysis—organizational level, culture, and leadership—are crucial in Greek Public Sector. Furthermore, statistical analysis confirmed that these three contents influence the leadership style applied.

More specific, this research has shown that the employees' perspective on conflict management techniques and leadership styles applied by the leader in the public sector varies depending on the position of responsibility. In addition, the pivotal role of collaborative leadership in the application of constructive conflict management techniques was highlighted.

Supervisors believe that the leader applies the techniques of compromise, confrontation, and organizing to deal with conflicts, while employees believe that the leaders avoid solving problems [51].

Supervisors recognize more than employees the application of collaborative leadership which is expressed through vision, belief in values, persuasion, team spirit, employee empowerment, positive communication and empathy. At the same time, supervisors believe more that the leader applies transformational leadership, recognizing that he is communicative, that he encourages and empowers employees and improves their work performance. Employees recognized more authoritarian leadership than the leader which is expressed through strictness, authoritarian behavior and obsessions [70].

Moreover, the study revealed a relation between collaborative and transformational leadership with compromising conflict management techniques, a finding in line with existing literature findings and in Samanta and Lamprakis research survey (2018). Thus, the results of the present study show the difference in perceptions between employees and supervisors. Accordingly, the study by Goldberg, Riordan & Zhang (2008) indicated that the most senior employees showed positive perceptions of their leaders in contrast to middle-level employees. In contrast, the study by Kanavas & Triantari et al. (2024) who examined leadership strategies and their influence on employee communication at all levels of the organization, argued that employees are mostly satisfied with the communication, leadership style and relationship with their leaders [14,95–97].

The results of this research also highlighted how leadership strategies and styles shape the organizational climate leading to a positive relationship between a leader's good communication with subordinates and organizational success. Also, the study by Kottke & Pelletier (2013) indicated that employees' perceptions of both leaders and their supervisors were significantly related to leadership style and climate [98].

Collaborative leadership was identified as the most important style for dealing with conflict. A collaborative approach to conflict leads to conflict effectiveness which in turn results in effective organizational performance. Leaders who are more cooperative are expected to apply more of the technique of compromise, confrontation and organizing to deal with conflicts [99].

Accordingly, Saeed et al. (2014) examined the relationship between leadership styles and conflict management style among leaders while handling interpersonal conflicts, concluding that collaborative leaders adopt more the technique of compromise and "organize" to deal with conflicts. According to Chan et al. (2008) the main characteristics of the cooperative leadership style are open communication, expression of care and respect for subordinates. These traits fit well with the compromise, confrontation, and conflict management techniques that focus on high concern and respect for others [29,30].

The same authors argue that it is possible that leaders who use collaborative conflict management styles enhance the social exchange process and gain trust from subordinates.

According to Triantari (2020) the collegial or cooperative type of leader seems to follow the Aristotelian view by focusing on the development of bonds of friendship between employees within the work environment, enhancing the cohesive bond between them. Of course, there is always the risk that the friendship between the leader and the team members will lead to mediocre performance and the relaxation of the business orientation, when the leader will not be able to convince or impose himself on the employees [31,32].

5.2. Implications

The findings of this research have significant implications, as studying conflict management and leadership styles within a specific context provides valuable information about it, regarding the cultural background and the professional field. In a case study approach, there is a better possibility for an in-depth analysis as it deals with real-life practices and challenges. This type of approach provides a deeper understanding of the dynamics involved. Moreover, linking specific techniques (e.g., compromise, confrontation) with certain leadership styles (e.g., collaborative, transformational) offers information about the best practices in organizational leadership, allows an enhanced conflict management analysis and suggests new models to deal with conflict resolution [21,100].

In particular, the findings of this research confirm a relation between collaborative and transformational leadership with constructive conflict management techniques. They also confirm differences in perspectives regarding leadership styles and conflict management techniques employed by leaders, suggesting a lack of communication and inability of leaders to address their message effectively to employees. Therefore, it casts a new light on the way cultural values influence conflict management as this incapability derives from the incompatibility of how the two groups (leaders/employees) perceive the world [101].

Moreover, the research contributes to the literature by revealing how cultural and structural factors in the Greek public sector impact leadership and conflict management and proves the importance of integrating these factors into leadership and conflict resolution theories, especially in hierarchical and bureaucratic environments [5].

Additionally, this research was conducted within a dynamic sector of the Greek Public Administration, where no previous similar studies existed. By introducing the concept of collaborative leadership and constructive management techniques, it initiates a new theoretical discussion on conflict resolution within complex public sectors. As leaders play a critical role in conflict resolution in these bureaucratic hierarchical contexts the study suggests a clear framework to enhance conflict resolution in public administration [102].

The findings regarding the correlation between different leadership styles and the use of conflict management techniques have practical implications. They provide recommendations based on evidence in order to improve leadership effectiveness in environments prone to conflicts. Departments of Public Works in Central Macedonia, Greece are susceptible to conflicts as they face bureaucratic challenges, resource constraints and conflicts of interest.

The study also reveals the tendency among leaders to favor authoritarian style and avoidance strategies according to employees' opinion, despite leaders believe that they are adopting collaborative and transformational approaches. That suggests that there is a disconnect in communication which can lead to unproductive conflicts dissatisfaction on the part of employees. Getting these perception differences aligned is crucial because the productivity and climate of an organization are influenced by the leadership style and how conflicts are handled by the leaders.

Policies could be implemented in the public sector introducing lifelong learning mandatory training programs for the leaders, not only concerning the theory about leadership but also providing the tools to enhance employees' engagement and commitment. Moreover, policies that allow leaders to be flexible and innovative regardless the strict

bureaucratic environment could foster trust and collaboration between employees and would improve leadership's effectiveness. Performance policies could include assessing leadership styles and conflict management practices as performance criteria, that could also act as indicators of the organization's effectiveness.

Such policies must link leadership practices with employees' needs and encourage employees to participate in decision making promoting accountability. The establishment of a formal conflict management resolution process which involves both supervisors and employees can improve understanding and operational efficiency. Similarly, Jung et al., (2020) argue that when the strategic management approach is used to empower employees, this factor can influence the performance in the firm [103].

Therefore, the study's findings give us the opportunity for targeted interventions to address to these specific perceptual gaps. Supervisors can be trained to improve their communication and empathy skills in order to better engage employees in the resolution process. The emphasis in training programs should be on collaborative and transformational leadership styles and on the development of relevant skills. This training could involve conflict resolution and negotiation tactics and mediation simulation. Moreover, mentoring and leadership coaching could be used as tools to teach leaders how to inspire, and motivate employees. Additionally, is crucial to empower employees to participate in the resolution process and to develop trust in their leaders by creating open lines of communication and feedback mechanisms. Thus, the tailored training programs addressed to supervisors could focus on effective communication, active listening, empathy and strategies to create an open dialogue environment. Leaders need to foster environments that prioritize collaboration, empowering employees to engage in conflict resolution actively [104]. Moreover, training programs addressing to the employees could emphasize in active participation, communication skills, soft and digital skills and conflict resolution strategies [105].

The insights from this research in Greek Public Sector may apply to similar bureaucratic environments worldwide as they offer a framework for examining perceptual gaps in leadership and conflict management in different cultural or organizational contexts. Besides, conflict management research is updating our understanding of what it takes to be an effective leader and how to develop it. Moreover, recommendations for changes that derive from these findings could improve workplace harmony and increase productivity, not only within the Greek public sector but also in a wider perspective [97].

5.3. Limitations-Suggestions

Despite its contribution the study has several limitations. The research was conducted in a specific context, Public Works of the Region of Central Macedonia in Greece and the results refer to employees of the public sector, with the majority being 40–59 years old, with a tertiary or postgraduate level of education and more than 10 years of service. The sample size was slightly less than adequate to ensure the power of the statistical tests conducted (J. Cohen, 2013). Using GPower to conduct the Mann Whitney test with a sample ratio of 2.5, $\alpha = 0.05$, power of 0.95 and an effect size of 0.5 required a size of 224 individuals (64 supervisors, 160 employees) when the sample of research were 186 individuals (54 supervisors and 132 employees). This sample size may pose limitations regarding the generalization of the study. The validation of the leadership questionnaire was performed on a sample smaller than the required 300. Therefore, generalization of the study is applicable to population with similar characteristics to the sample. On the other hand, the correlation research offers the possibility to generate research for further experimental studies [90].

The use of convenience sampling poses challenges to external validity. There are several biases that can affect the generalization of the study like the non-random selection of the sample which make the sample non-representative or that certain groups with similar characteristics are overrepresented.

The statistical significance can be enhanced by future researchers by larger and more diverse sample on random selection of public sector employees addressing Public Works' services throughout the country. Moreover, as the results of the research derive from a single sector of public services (Public Works), it would benefit generalization if similar research could be carried out in other public sectors (e.g., Security Forces, Education, Taxation, Health Services).

Furthermore, limitations exist also due to the cultural influences that derive of the bureaucratic and hierarchical structure of Greek public sector. Thus, cultural values as high-power distance and collectivism that are prevalent in Greek public sector may not align with other countries. Therefore, cross-cultural studies collecting data from different countries would provide valuable insights regarding the applicability of these results across different context.

Finally, the data for this study were exclusively collected from the Public Works Departments within the Greek public sector, specifically focusing on the differences in organizational levels between Municipalities (first level of local authority) and Prefectures (second level of local authority). While these departments belong to different organizational levels, the responses from employees remained consistent across both levels. This consistency reinforces the validity of the results and ensures that the differences in organizational levels do not introduce significant variability in the findings. The statistical analysis provides clear and transparent patterns that strongly support the research conclusions, making the relationship between organizational levels, leadership styles, and conflict management techniques evident. Therefore, the credibility of the conclusions and the management implications drawn from this study remains robust and convincing to readers.

6. Conclusions

Comparative perspective is not commonly explored in leadership literature; thus, the results of this study provide fresh insights into the difference in perceptions between hierarchical roles (employees and supervisors). Supervisors supported more than employees that the leader employs the conflict management techniques of compromise, confrontation and organizing and that he/she applies collaborative and transformational leadership style. In contrast, employees more strongly supported the leader's avoidance technique and authoritarian leadership. The revealing of this gap, offers the opportunity to introduce policies and trainings to address it. These policies and trainings should incorporate sustainability principles to ensure long-term improvements in workplace productivity and harmony. An impactful initiative it could be the creation of a Leadership Academy under the National Center of Public Administration and Local Government (EKDDA), where structured leadership programs will be offered in public officials as there isn't such an institution in Greece. There is a major digital transformation that is taking place in Greek Public Administration now and it is a great opportunity to incorporate digital and agile training courses for public leaders with the use of new technology tools.

The research has significant contribution by revealing the relation between leadership styles and conflict management techniques within the Greek Public Sector. By integrating sustainability into leadership practices, organizations can achieve not only immediate conflict resolution but also sustained organizational effectiveness. The finding that collaborative leadership is expected to lead to more frequent implementation of a constructive conflict management technique introduces a new perspective on conflict resolution in a bu-

reaucratical, hierarchical environment like the Greek Public Sector. The research advances academic understanding but also offers practical guidance for developing tailored strategies in public sector management. Furthermore, the research initiates a discussion on the applicability of the findings in different cultural environments, enhancing the understanding of sustainable workplace dynamics.

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